United States Naval Infrastructure Preservation Command

REVOLUTIONARY TECHNOLOGY CO. // SYSTEM MANIFESTS SERIES

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Subsystem: Hard-Deterministic Multi-Variable Marine Co-Processor

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NAVAL OPERATOR PRE-SAIL FIELD TRAINING GUIDE

Re-Animating Hardened Base Facilities & Tactical Weapon Systems via the Unified UNIVAC Co-Processor Plant

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1. Functional Overview & Security Scope

This manual serves as the standard operating procedure (SOP) for field engineers and system operators tasked with re-animating, validating, and managing hardened infrastructure controlled by legacy UNIVAC architectures.

When a ship anchors or hooks into a shore-side base network link, this co-processor architecture acts as an immediate **Dynamic Interface Alignment Engine** [INDEX]. It completely isolates high-overhead administrative computing logs into an independent background processing ring while locking down safety-critical control laws, weapon alignments, and heavy utility valves inside a hard-deterministic 50Hz main timing loop [INDEX].

All operators must execute the following procedures precisely to clear the hardware safety interlocks, align dynamic device network identities, and verify system compliance flags before departure.

```
       [ HARDWARE POWER ISOLATION & RS-422 STEP-RESPONSE CHECK ]
                              |
                              ▼
         [ PRE-FLIGHT BLOCK-CHAIN LEDGER SCAN & SCHEMAS ]
                              |
                              ▼
         [ DYNAMIC MULTI-MODEL BIT-WIDTH SELECTION (HAL) ]
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                          |
                          ▼
       [ RAW PACKET TACTICAL NETWORK ADAPTATION LOOP ]
                          |
                          ▼
     [ ENGAGE WATCHDOG & SECURE 50Hz MAIN CORE AUTO-LOOPS ]
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2. Phase 1: Hardware Isolation & Voltage Balance Check (Pier-Side)

Before initializing any software loops, you must verify the physical integrity of the electrical differential communication cables and shoreside transformers.

- **[ ] Action 1.1: Power Grid Interlock Verification**
  Locate the main 24V DC input breaker feeding the local edge co-processor enclosure. Set the toggle switch to OFF. Disconnect the external shoreside power umbilical line.
- **[ ] Action 1.2: RS-422 Differential Breakout Audit**
  Attach a digital multimeter to the differential breakout pins on the main RS-422 interface terminal strip (/dev/ttyUSB4 or COM7) [INDEX].
- **[ ] Action 1.3: Zero-Point Cross-Talk Evaluation**
  With the vessel sitting flat at the pier (0.0° roll, 0.0 rad/s roll rate), record the differential voltage between the Port (Pin 3/4) and Starboard (Pin 7/8) loops [INDEX].
- **[ ] Evaluation Rule**: The cross-talk value must measure exactly:
  $$\Delta V_{\text{differential}} = 0.00\,\text{VDC} \pm 15\,\text{mV}$$
  If the variance exceeds 15mV, you must manually adjust the amplifier trim-pots to null out the DC line drift before proceeding.

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3. Phase 2: Pre-Flight Blockchain Ledger Verification

To comply with international maritime regulations and federal anti-corruption guidelines, the software refuses to clear steering loops if historical logs show evidence of manual modification or tampering [INDEX].

- **[ ] Action 2.1: Execute Pre-Flight Compliance Check**
  Power on the co-processor enclosure. Open an elevated system terminal and invoke the validation script:
  `python /src/config/boot_verification_suite.py`
- **[ ] Action 2.2: Verify Crypto Hash Chain Integrity**
  The script re-calculates the 8-bit XOR checksum parameters and scans the **Flag Halyard Records**, **MARPOL Bilge Logs**, **Weapon/Compass Telemetry Records**, and **Shore Facility Ledgers** row-by-row [INDEX].
- **[ ] Mathematical Verification Formula**: The validator asserts that every line matches the

block-chained SHA-256 signature anchor:
$$\mathcal{H}_{n} = \text{SHA256}\Big(\text{Metrics}_{body} \,\Vert{}\, \mathcal{H}_{n-1}\Big)$$

- **[ ] Evaluation Rule**: If a single byte of historical crane, valve, or weapon data was altered, the check fails. The screen will flash a critical alarm banner block across the operator console, and startup will immediately abort.

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4. Phase 3: Dynamic Multi-Model Bit-Width Ingestion Selection

Because different base facilities and vintage warships use completely incompatible bit architectures, you must explicitly match the co-processor's masking registers to the target UNIVAC computer [INDEX].

- **[ ] Action 3.1: Identify Target Computer Generation**
  Consult the facility architecture log. Locate the master model configuration line inside manifest.json.
- **[ ] Action 3.2: Select System Word-Packing Profile**
  Toggle your system word-packing selection using one of the eight hardcoded bitmask arrays to filter register bleed:$$\begin{aligned} \text{UNIVAC 1219 (18-Bit Auxiliaries):} \quad &\mathbf{W}_{\text{clean}} = \mathbf{W}_{\text{raw}} \ \ \& \ \ \mathtt{0x0003FFFF} \\ \text{AN/UYK-43 (32-Bit Aegis Core):} \quad &\mathbf{W}_{\text{clean}} = \mathbf{W}_{\text{raw}} \ \ \& \ \ \mathtt{0xFFFFFFFF} \\ \text{AN/UYK-20 \& 14 (16-Bit Minicomputer):} \quad &\mathbf{W}_{\text{clean}} = \mathbf{W}_{\text{raw}} \ \ \& \ \ \mathtt{0x0000FFFF} \\ \text{UNIVAC 490 \& 494 (30-Bit Real-Time):} \quad &\mathbf{W}_{\text{clean}} = \mathbf{W}_{\text{raw}} \ \ \& \ \ \mathtt{0x3FFFFFFF} \\ \text{UNIVAC 1107 \& 1108 (36-Bit Scientific):} \quad &\mathbf{W}_{\text{clean}} = \mathbf{W}_{\text{raw}} \ \ \& \ \ \mathtt{0xFFAAAAAA} \end{aligned}$$

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5. Phase 4: Tactical Network Identity Alignment Loop

To communicate with legacy networks, the co-processor must align its physical network interface identity on the fly to match the exact hardware address signature of the target asset.

- **[ ] Action 4.1: Secure Physical Ethernet Loopback**
  Verify the primary network interface card is patched directly into an unmanaged, low-latency gigabit network switchboard to establish an active loopback link.
- **[ ] Action 4.2: Run Elevated Handshake Test**
  Execute the raw network packet validation script using elevated privileges to access raw network sockets:
  ```
  sudo python /src/config/hardware_network_validator.py
  ```
- **[ ] Action 4.3: Validate Latch-Bit Latching Accuracy**
  The system reads your target vessel hull class, builds a raw 6-byte Ethernet frame, and verifies the hardware pin matrix variables inside Octet 6:

$$\text{Octet 6} = (\text{Bit 2} \ll 2) \mid (\text{Bit 1} \ll 1) \mid \text{Bit 0}$$

- **[ ] Evaluation Rule**: Verify that **Bit 2 (Verification Latch)** exactly duplicates **Bit 0 (Peripheral Device Index)**. If the bits do not match, the link will time out. The system will lock out the network bus to block unauthorized data traffic [INDEX].

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6. Phase 5: Hardware Watchdog Engagement & Final Launch

Once all hardware, cryptographic, and network identities return a clean pass signature, you are cleared to engage the main automation loops.

- **[ ] Action 5.1: Clear Test Benches**
  Remove all loopback plugs from your serial terminal strips. Reconnect your physical external sensor feeds (True Heading Compass, Echo Sounder, Gun-Ring Encoders).
- **[ ] Action 5.2: Fire the Master Core Autopilot**
  Execute the master bootloader orchestration script:
  ```
  python /src/main.py
  ```
- **[ ] Action 5.3: Monitor Watchdog Heartbeats**
  Confirm via your flat-panel Tkinter dashboard console that the **Asynchronous Multi-Ledger Hardware Watchdog** is actively tracking timestamps [INDEX].
- **[ ] Safety Interlock Engagement**: If a file append blocks or a serial line disconnects during operations, the watchdog will trip within 1.5 seconds, automatically forcing motor torque to a safe zero baseline (0.0 Nm) and locking the anchor windlass brakes to secure the vessel [INDEX].

PRE-SAIL ALIGNMENT STATUS: APPROVED

Certified Compliance Officer: Dr. Correo Hofstad & Dr. Jensen Huang
Master Operations Key: UNIVAC-RECLAMATION-COMPLIANT-2026

The platform is officially verified, cryptographically chained, and certified for active operational transit or museum installation display [INDEX].